

Dylan, 17 · helmet-to-helmet hit at football practice · brief unconsciousness · no memory of the play · headache, nausea, photophobia · normal CT · slow on cognitive testing · **sport-related concussion (mild TBI)**

BIO 004 · WEEK 7 · CASE DEEP DIVE

Gross Anatomy & Neuronal Integration

A high-school athlete with a hit to the helmet, no memory of the play, and a brain that needs to come offline before it heals.

PATIENT CASE

Dylan is 17, a high-school junior playing varsity football. During practice he took a **helmet-to-helmet hit during a tackle**. He was briefly unconscious for about 10 seconds, was helped to the sideline confused, and now can't recall the play.

On exam he has a **headache, mild nausea, sensitivity to light**, slow reaction time, difficulty concentrating. Normal cranial nerves, normal strength, normal coordination. CT of the head is normal. SCAT (sideline assessment) is abnormal. The diagnosis is **concussion (mild TBI)**. He is removed from play and started on a graded return-to-learn and return-to-play protocol.

Concussion is a functional injury to the brain without obvious anatomic damage on standard imaging. To understand what's happening to Dylan's neurons, you need to know how the brain is organized, how reflex arcs work, and how sensory and motor pathways tie central integration to peripheral function.

GUIDING QUESTION

What are the major divisions of the brain, what is a reflex arc, and how do sensory and motor pathways connect periphery to center?

PREWORK

Companion to your Week 7 Gross Anatomy & Neuronal Integration prework page.

Six divisions, embryologically organized

Cerebrum

Largest. Left and right hemispheres connected by corpus callosum. Cortex (gray matter, cognition, sensation, voluntary movement), deep nuclei (basal ganglia), white matter tracts.

Diencephalon

Thalamus (sensory relay), hypothalamus (homeostasis, endocrine), epithalamus (pineal).

Midbrain

Upper brainstem. Eye movement control (CN III, IV), auditory and visual reflexes.

Pons

Middle brainstem. Relays between cerebrum and cerebellum. Cranial nerves V-VIII.

Medulla oblongata

Lower brainstem. Cardiac, respiratory, vomiting, swallowing centers. Cranial nerves IX-XII.

Cerebellum

Coordinates movement, balance, motor learning. Receives input from cortex and from proprioception.

CSF-filled spaces inside the brain

Four ventricles: two **lateral** (one in each cerebral hemisphere), **third** (midline diencephalon), **fourth** (between brainstem and cerebellum). Connected by narrow passages. CSF is produced by choroid plexus inside each ventricle, flows through them, exits the 4th ventricle into the subarachnoid space, circulates around brain and spinal cord, and is absorbed back into venous blood by arachnoid granulations. Covered in detail on the CNS Meninges & CSF page.

Stimulus to response without thinking

A **reflex arc** is the simplest functional circuit in the nervous system. Five components:

1. **Receptor** detects stimulus (e.g., muscle spindle senses stretch).
2. **Sensory neuron** carries signal into the CNS.
3. **Integration center** (spinal cord or brainstem). Monosynaptic = one synapse (stretch reflex).
Polysynaptic = multiple synapses.
4. **Motor neuron** carries signal out.
5. **Effector** (muscle, gland) responds.

Classic example: the *patellar (knee-jerk) reflex*. Strike patellar tendon → quadriceps stretches → muscle spindle fires → sensory neuron enters L2-L4 cord → directly synapses on alpha motor neuron → quadriceps contracts → leg extends. Monosynaptic, doesn't require brain involvement.

Three-neuron chain from periphery to cortex

Most somatic sensation reaches the cortex via three neurons in series:



First-order neuron

From receptor in periphery to spinal cord (or brainstem). Cell body in dorsal root ganglion.



Second-order neuron

Crosses to the opposite side. Ascends to thalamus.



Third-order neuron

From thalamus to primary sensory cortex (postcentral gyrus).

Two main pathways: **dorsal column-medial lemniscus** (fine touch, proprioception, vibration; crosses in medulla) and **spinothalamic** (pain, temperature, crude touch; crosses at the level of entry into the cord). The crossing pattern explains why a left-sided spinal cord lesion can cause one type of loss on one side and another type on the other.

Two-neuron chain from cortex to muscle

Voluntary movement uses the **corticospinal tract**:

Upper motor neuron

Cell body in primary motor cortex (precentral gyrus). Axon descends through internal capsule, brainstem, crosses at the pyramidal decussation in the medulla, descends in spinal cord lateral corticospinal tract.

Lower motor neuron

Cell body in spinal cord anterior horn (or brainstem for CN nuclei). Axon exits via spinal/cranial nerve to skeletal muscle.

Lesions of upper motor neurons cause weakness with *hyperreflexia*, *spasticity*, *positive Babinski*. Lesions of lower motor neurons cause weakness with *hyporeflexia*, *atrophy*, *fasciculations*.

Why a normal CT doesn't mean a normal brain

Concussion is a **diffuse, transient neurometabolic disturbance** after rotational/acceleration injury. Mechanical force stretches axons, disrupting cell membranes. Massive disordered ion fluxes (K^+ out, Ca^{++} in, glutamate release). The cell membrane tries to restore equilibrium, but Na/K-ATPase pumps run out of ATP. The brain becomes hypermetabolic but with impaired oxygen and glucose delivery (mismatch). Function is impaired even though no gross structural damage is visible on CT.

Why Dylan needs to come offline. The recovering brain is in a state of *metabolic vulnerability*: even a second mild hit during this period can produce catastrophic injury (*second-impact syndrome*). Cognitive rest plus graded return-to-activity (return-to-learn before return-to-play) is the standard.



DRAW ZONE 1 -- *Diagram the brain showing the six major divisions: cerebrum, diencephalon, midbrain, pons, medulla, cerebellum.*

Mark the cerebral lobes (frontal, parietal, temporal, occipital) and the brainstem boundaries.



DRAW ZONE 2 -- Sketch a simple monosynaptic reflex arc (patellar) and a polysynaptic withdrawal reflex. Label all five components.

For the withdrawal reflex, include the inhibitory interneuron that relaxes the antagonist (crossed extensor).

A functional injury in a structurally intact brain

Concussion is a teaching case in how the nervous system can be disrupted at the cellular level without anything showing on scans.

"Brief loss of consciousness"

Diffuse axonal stretch transiently disrupts ascending reticular activating system function. Most concussions don't involve LOC; when they do, it's brief.

"Amnesia for the event"

Hippocampal memory consolidation is interrupted. Memories not yet consolidated are lost; new memories may be impaired for hours-days.

"Headache, photophobia, nausea"

Disordered neurotransmitter release and altered cerebral blood flow. The same triad seen in migraine — concussion physiology overlaps with migraine physiology.

"Slow reaction time, difficulty concentrating"

The metabolic mismatch impairs neuronal function broadly. Cognitive demand exceeds available energy supply temporarily.

"Normal CT"

CT detects gross structural injury (hemorrhage, large contusion, skull fracture). Concussion is a microscopic, functional injury. Most concussions have normal imaging.

"Graded return-to-play"

5-stage protocol: symptom-free at rest → light aerobic → sport-specific exercise → non-contact training → full contact practice → return to play. At least 24 hours between stages. Any symptom recurrence drops back a stage.

Cerebrum, reflex, pathway, integration. Dylan's brain is intact on every structural test but disrupted at the neuronal level.

